

Foundation of Programming

Assignment No. 13

PROBLEM STATEMENT:

Write a program in C to accept a number from the user and print the digits of the number in reverse order.

OBJECTIVE:

1. To understand number manipulation using loops
2. To learn the use of modulo (%) and division (/) operators.
3. To implement logic for reversing digits of a number.
4. To improve problem-solving skills using iterative structures.

THEORY:

A program that reverses the order of numbers works by taking a sequence of values and rearranging them so that the first element becomes the last, the second becomes the second-last, and so on until the sequence is completely inverted.

ALGORITHM:

STEP 1:	START
STEP 2:	Declare variables n, temp, d = 0, r = 0
STEP 3:	Accept the value of n from the user
STEP 4:	Run a loop till temp != 0 Store the remainder of temp divided by 10 in d Store the sum of r * 10 and d in r temp = temp / 10
STEP 5:	Display r
STEP 6:	STOP

PLATFORM : Linux

CODE:

```

1  #include <stdio.h>
2
3  int main()
4  {
5      printf("\nEnter a number\n");
6      int n = 0;
7      scanf("%d", &n);
8
9      int temp = n, d = 0, r = 0;
10     while (temp != 0)
11     {
12         d = temp % 10;
13         r = r * 10 + d;
14         temp = temp / 10;
15     }
16
17     printf("\nReversed Number = %d\n", r);
18     return 0;
19 }

```

OUTPUT:

```

Enter a number
87

Reversed Number = 78

```

CONCLUSION:

Thus, the program to print the digits of a number in reverse order was successfully implemented using C programming. The program demonstrates the effective use of loops, modulo operator, and division operator.